End Sem Exam Prob + Stat (MCC 505) 2023-2024

1 (a) Given $n = 50$, $\Sigma x = 212$, $\Sigma x^2 = 902.8$

$\Sigma y = 261$, $\Sigma y^2 = 1457.6$

$$\sigma_x^2 = \frac{1}{n}\Sigma x^2 - \bar{x}^2 = \frac{1}{50}(902.8) - \left(\frac{212}{50}\right)^2$$

$$= 18.056 - 17.9776 = 0.0784$$

$\sigma_x = 0.28$, $\bar{x} = \frac{212}{50} = 4.24$, $CV_x = \frac{\sigma_x}{\bar{x}} = \frac{0.28}{4.24} = 0.06604$ $\left(\frac{3}{4}\text{ mark}\right)$

$$\sigma_y^2 = \frac{1}{n}\Sigma y^2 - \bar{y}^2 = \frac{1}{50}(1457.6) - \left(\frac{261}{50}\right)^2 = 29.152 - 27.2484$$

$= 1.9036$, $\sigma_y = 1.3797$, $\bar{y} = 5.22$

$CV_y = \frac{1.3797}{5.22} = 0.2643$ $\left(\frac{3}{4}\text{ mark}\right)$

$CV_x < CV_y$, Hence length X is more consistent $\left(\frac{1}{2}\text{ mark}\right)$

(b) Given $\mu_1' = \bar{x} = 5$, $\mu_2 = 20$, $\mu_3 = 140$

$$\mu_1'(A) = \frac{1}{N}\sum_{i=1}^{n} f_i(x_i - A) = \frac{1}{N}\sum_{i=1}^{n} f_i x_i - A = \bar{x} - A$$

Hence, $\mu_1'(10) = \bar{x} - 10 = 5 - 10 = -5$

$$\mu_3'(10) = \mu_3 + 3\mu_2\mu_1'(10) + (\mu_1'(10))^3$$

$$= 140 + 3 \times 20 \times (-5) + (-5)^3 = 140 - 300 - 125$$

$$= -285$$

Hence $\boxed{\mu_3'(10) = -285}$ (2 marks)

2 (a) $\beta_2 \geq 1$

We known $\beta_2 = \frac{\mu_4}{\mu_2^2}$ $\Rightarrow$ we have to prove $\frac{\mu_4}{\mu_2^2} \geq 1$

$\Rightarrow \mu_4 \geq \mu_2^2$ or $\mu_4 - \mu_2^2 \geq 0$

$$= \frac{1}{N}\sum_{i=1}^{n} f_i(x_i - \bar{x})^4 - \left(\frac{1}{N}\sum_{i=1}^{n} f_i(x_i - \bar{x})^2\right)^2 \geq 0$$

Let us assume $z_i = (x_i - \bar{x})^2$

which gives $\frac{1}{N}\sum_{i=1}^{n} f_i z_i^2 - \left(\frac{1}{N}\sum_{i=1}^{n} f_i z_i\right)^2 \geq 0$

$= \sigma_z^2 \geq 0$ which is always true.

Hence, $\beta_2 \geq 1$ (3 marks)

(b) $U = x - 6$, $\Sigma f = 300$, $\Sigma fU = -61$, $\Sigma fU^2 = 1445$

$\Sigma fU^3 = -1285$, $\Sigma fU^4 = 16469$

Soln. If $U = \frac{X-A}{h}$, $\mu_{rx} = h^r \mu_{ru}$

Here $A = 6$, $h = 1$, hence $\mu_{rx} = \mu_{ru}$

Hence $\mu'_{1u} = \frac{-61}{300} = -0.2033$, $\mu'_{2u} = \frac{1}{300}(1445) = 4.8167$

$\mu'_{3u} = \frac{1}{300}(-1285) = -4.2833$, $\mu'_{4u} = \frac{1}{300}(16469)$
$= 54.8966$

$\mu_{1u} = 0$ (always), $\mu_{2u} = \mu'_{2u} - \mu'^2_{1u} = 4.8167 - (-0.2033)^2$
$= 4.7754$

$\mu_{3u} = \mu'_{3u} - 3\mu'_{2u}\mu'_{1u} + 2\mu'^3_{1u} = -1.3624$

$\mu_{4u} = \mu'_{4u} - 4\mu'_{3u}\mu'_{1u} + 6\mu'_{2u}\mu'^2_{1u} - 3\mu'^4_{1u} = 52.6028$

Here, $\mu_{rx} = \mu_{ru}$

Hence $\mu_{1x} = 0$, $\mu_{2x} = 4.7754$, $\mu_{3x} = -1.3624$
$\mu_{4x} = 52.6028$

$$\beta_1 = \frac{\mu_{3x}^2}{\mu_{2x}^3} = \frac{(-1.3624)^2}{(4.7754)^3} = 0.0170$$ (-vly skewed as $\mu_{3x} < 0$)

$$\beta_2 = \frac{\mu_{4x}}{\mu_{2x}^2} = \frac{52.6028}{(4.7754)^2} = 2.3067$$ (Platykurtic)

$\mu_{1x}, \mu_{2x}, \mu_{3x}, \mu_{4x}$ $\left(\frac{1}{2} + \frac{1}{2} + 1 + 2 = 4 \text{ marks}\right)$

β_1, β_2 & comments $\left(\frac{1}{2} + \frac{1}{2} = 1 \text{ mark}\right)$